

C.W

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Design a logic circuit that has been three inputs A, B, C and where output will be High when a majority of inputs is High.

Ans:

A	B	C	output (y)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$Y = \bar{A}BC + A\bar{B}C + A\bar{B}\bar{C} + ABC$$

$$= BC(\bar{A} + A) + A\bar{B}C + A\bar{B}\bar{C}$$

$$= BC + A\bar{B}C + A\bar{B}\bar{C}$$

$$= C(B + A\bar{B}) + A\bar{B}\bar{C}$$

$$= C(B + \bar{B})(B + A) + A\bar{B}\bar{C}$$

$$= BC + AC + ABC$$

$$= BC + A(C + BC)$$

$$= BC + A(C + \bar{C})(C + B)$$

$$= BC + AC + AB$$

$$= AB + BC + CA$$

Answer:

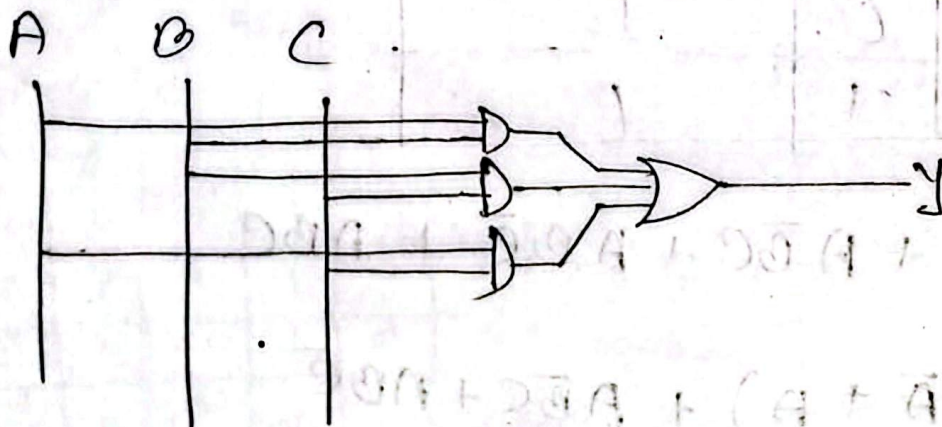
$$Y = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

$$= \bar{A}BC + ABC + A\bar{B}C + ABC + AB\bar{C} + ABC$$

$$= BC(\bar{A} + A) + AC(\bar{B} + B) + AB(\bar{C} + C)$$

$$= BC \cdot 1 + AC \cdot 1 + AB \cdot 1$$

$$= AB + BC + AC$$



$$Y = AB + BC + AC$$

HW: Same question, but just high and low.